Vertical Stabilizer – General Data

1. General

The vertical stabilizer is secured by four bolts at front and rear attachment fittings to corresponding fittings on the rear fuselage.

The vertical stabilizer comprises spars, ribs, and skins reinforced by stringers. The front and rear attachment fittings are riveted to the bottom of the front and rear spars. The top and center hinge plates for rudder attachment are riveted to the ribs. A detachable cap on top of the stabilizer houses the VOR antenna and the anti-collision light. On aircraft with lightning protection (Mod 6/1272), a horn assembly is attached to the cap, and protection straps and a conduit extension assembly are fitted between the cap and stabilizer structure. Structural provision has been made on each side of the vertical stabilizer for the installation of a half loop type VOR antenna (Mod 6/2003). On aircraft with Mod 6/2171 incorporated, two KMG 7010 magnetometers are mounted in the vertical stabilizer, one for each channel of the ADAHRS. (Refer to 34-20-00, Attitude and Direction – General Data).

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Vertical Stabilizer

1) Remove fairings enclosing vertical stabilizer and horizontal stabilizer to rear fuselage.

2) Remove rudder. (Refer to 27-20-11, Rudder – Maintenance Practices).

3) Disconnect electrical wiring at disconnect in forward end of vertical stabilizer.

4) Disconnect VOR antenna wiring in rear fuselage.

5) Disconnect ADAHRS magnetometers (Post Mod 6/2171).

6) Attach sling SD12507 to vertical stabilizer and take weight. (Refer to 07-00-00, Lifting and Shoring – General Data).

7) Remove four bolts, saddle washers and washers at front and rear attachment fittings of vertical stabilizer to rear fuselage.

8) Lift vertical stabilizer clear of aircraft. Remove four barrel nuts and barrel nut retainers from rear fuselage attachment fittings.

B. Install Vertical Stabilizer

1) Attach sling and position vertical stabilizer at front and rear attachment fittings on rear fuselage.

2) Install four bolts with saddle washers and washers at attachment fittings. Check that plain shank of each bolt is approximately flush with inside surface of fitting.

Note Maximum length of thread in fitting 0.10 inch. Maximum length of plain shank of bolt extending through fitting 0.10 inch. If necessary, add washers under existing washer on bolt.

3) Withdraw bolts sufficiently to install barrel nuts and barrel nut retainers. Tighten front and rear attachment bolts to 340 to 350 pound–inches.

4) Remove sling from stabilizer.

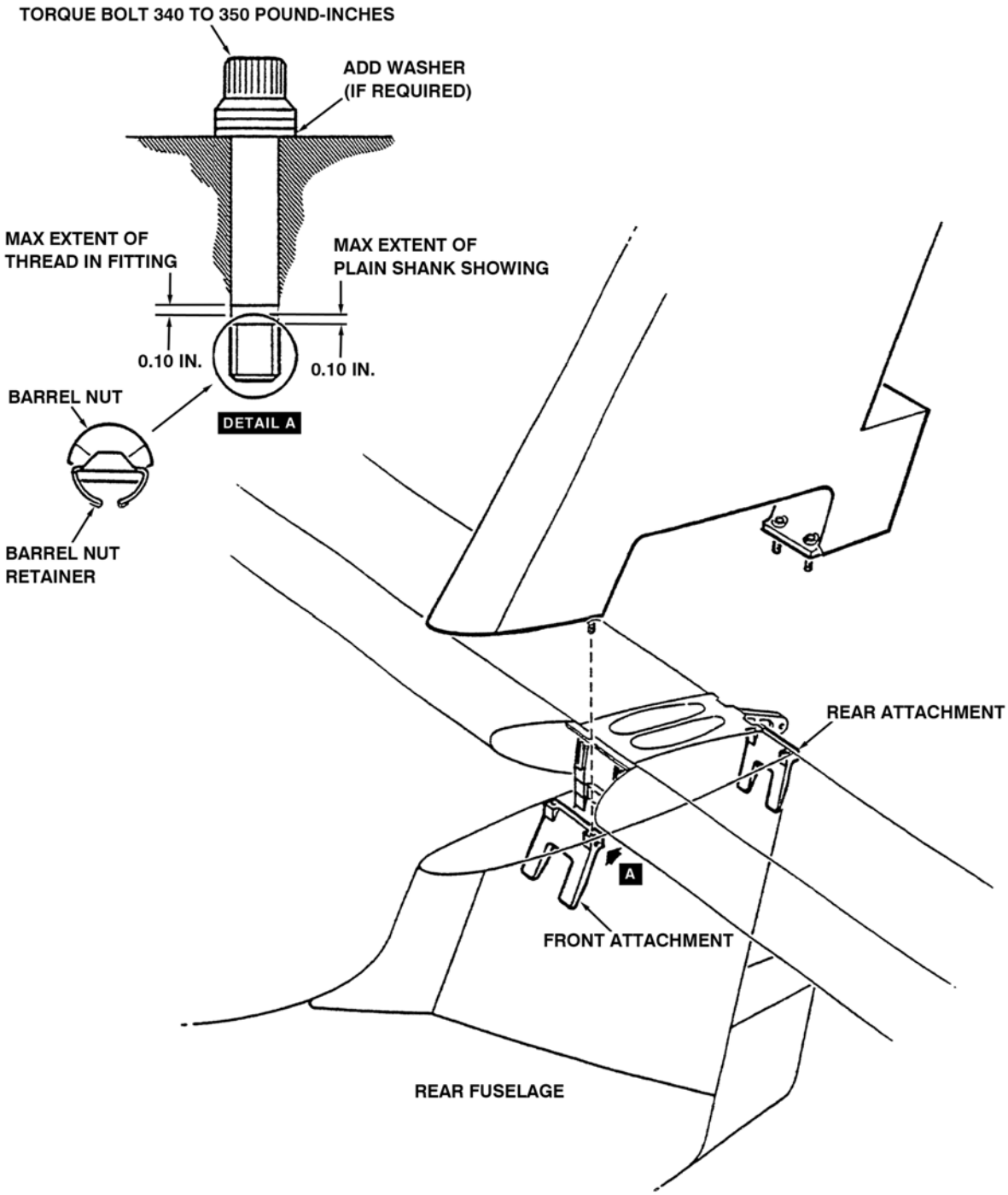
5) Connect VOR antenna wiring in rear fuselage.

6) Connect ADAHRS magnetometers (Post Mod 6/2171).

7) Connect electrical wiring at disconnect in forward end of vertical stabilizer.

8) Install rudder. (Refer to 27-20-11, Rudder – Maintenance Practices).

9) Install fairings enclosing vertical stabilizer and horizontal stabilizer to rear fuselage.



Vertical Stabilizer Installation
Figure 1